

A.Y. 2025-26

# Design of Secured FTTH for Routing and Wavelength Assignment (RWA) in Optical Networks

SDG MAPPING

PROJECT GROUP NO - 4

INTRODUCTION

Fiber to the Home (FTTH) has emerged as a key technology for delivering high-speed broadband connectivity through optical fiber networks. However, Routing and Wavelength Assignment (RWA) in optical networks faces challenges such as wavelength continuity constraints, signal contention, and security vulnerabilities at the physical layer.

Traditional WDM-based solutions improve bandwidth utilization but are often complex and costly. Visible Light Communication (VLC), which uses LEDs and laser sources for data transmission, offers a secure and low-cost alternative due to its directional and line-of-sight nature.

This project proposes a secure FTTH prototype integrating VLC and encryption techniques to simulate Routing and Wavelength Assignment in a controlled indoor environment..

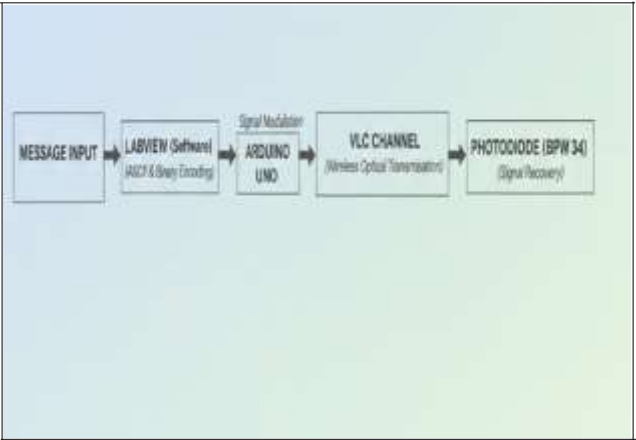
AIMS & OBJECTIVES

| Section     | Description                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aim         | To design and implement a secure FTTH prototype using Visible Light Communication (VLC) for simulating Routing and Wavelength Assignment (RWA) in optical networks. |
| Objective 1 | Develop laser-based string transmission by converting text into ASCII and binary format for optical communication.                                                  |
| Objective2  | Validate single-character transmission to ensure 100% symbol-level synchronization and bit accuracy.                                                                |
| Objective3  | Implement LED-based VLC communication to simulate a low-cost FTTH optical link.                                                                                     |
| Objective 4 | Integrate encryption and decryption mechanisms to enhance physical-layer security.                                                                                  |
| Objcclve 5  | Analyze system performance under indoor conditions based on accuracy, delay, and alignment sensitivity.                                                             |

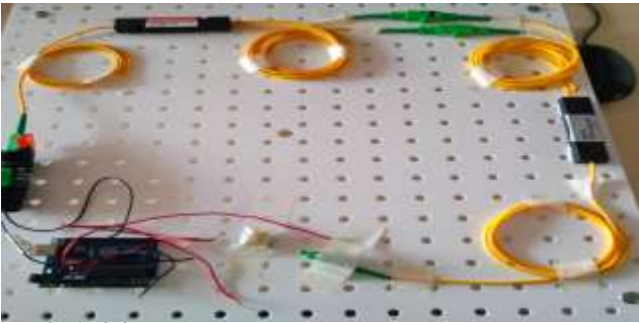
TECH STACK

| Stage                      | Details                                   | Tools/ Techniques               |
|----------------------------|-------------------------------------------|---------------------------------|
| Hardware setup             | Optical transmission & reception setup    | Arduino Uno, 660nm Laser, BPW34 |
| Data Processing            | Text to Binary encoding & encryption      | Lab VIEW                        |
| Communication              | VLC-based signal transmission & detection | LED/Laser, Op-Amp Circuit       |
| Component                  | Technology /Tools                         |                                 |
| Frontend                   | LabView Interface                         |                                 |
| Back-end / Core Processing | Arduino (C++ Programming)                 |                                 |
| Optical Channel            | Laser Diode / LED (VLC)                   |                                 |
| Receiver                   | BPW34 Photodiode + Op-Amp                 |                                 |
| Deployment                 | Arduino IDE                               |                                 |
| Encryption Module          | Substitution Cipher ( Lab View)           |                                 |

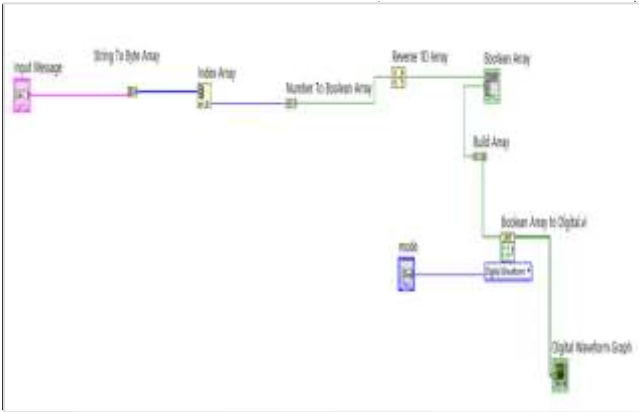
SYSTEM ARCHITECTURE



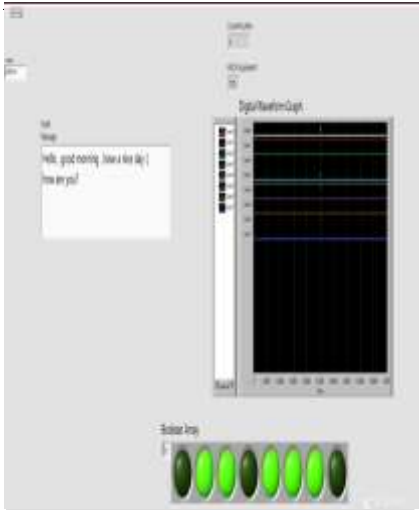
MODEL VALIDATION



PROJECT SNAPSHOTS



RESULTS



CONCLUSION

This project successfully implements a secure FTTH communication system to address the Routing and Wavelength Assignment (RWA) problem using Visible Light Communication (VLC). The integration of LabVIEW for encoding and monitoring, along with Arduino-based signal modulation and laser transmission, enabled reliable optical data transfer.

The experimental results confirm accurate message transmission through electrical-to-optical and optical-to-electrical conversion stages, demonstrating stable signal recovery using a photodiode and amplifier circuit.

The developed prototype highlights the feasibility of VLC as a secure, low-cost, and energy-efficient solution for next-generation optical access networks.

Furthermore, the system demonstrates that wavelength-based routing concepts can be effectively visualized and validated through a controlled VLC setup. The successful implementation of encoding, modulation, transmission, and decoding stages confirms the practical feasibility of the proposed architecture for secure optical communication.

LITERATURE SITES

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